GECKO USER MANUAL

Sensorimotor Neuroscience and Learning Laboratory – Pennsylvania State University

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Introduction:

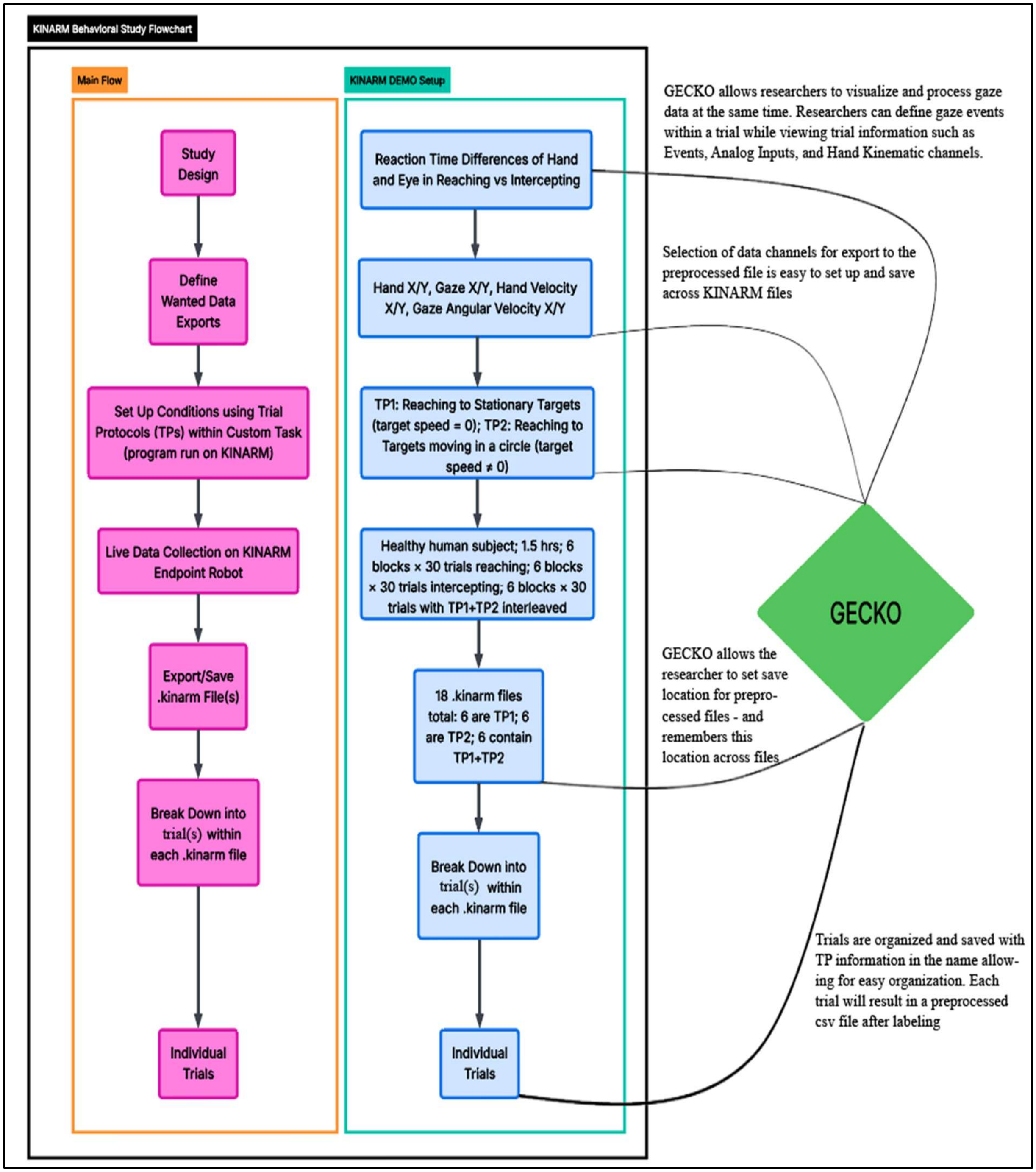


Figure 1. Flowchart of a typical behavioral study using a KINARM, alongside the same flow of an example paradigm. Lines and captions indicate areas where GECKO software is applicable.

To introduce GECKO, let's start in a familiar place – a KINARM behavioral study. GECKO is meant to be used with KINARM studies, specifically, it works only with **.kinarm files**. The flowchart above shows what a typical behavioral study using a

KINARM Endpoint might look like (pink boxes), and a ‘mock’ study fitted to that same flow (blue boxes). As seen on the right, GECKO is a software that applies to many fields of the KINARM study, not just the trial-by-trial analysis of gaze. It serves as the one-stop first pass method for pre-processing .kinarm files with gaze data. The figure above can be useful to help the user understand where and how GECKO will be implementable to their study.

This manual follows [demo_1.kinarm] from upload to export so each function of GECKO is introduced in the order a user would encounter it

The DEMO Data:

The data file [demo_1.kinarm] contains 6 trials of a simple KINARM task. Eye-tracking was done with an EyeLink 1000, and the task was performed on a KINARM Endpoint Robot. The participant puts their hand in a start position and waits (Data Logging begins from here). A target will appear to the left or right of their hand, and they are trying to reach the target with the right manipulandum. In the first 2 trials, the target will appear and remain stationary – and in other 4 it will start moving in a counterclockwise direction upon appearance (Trial Protocol 13 and 14 are the stationary targets on the right and left,

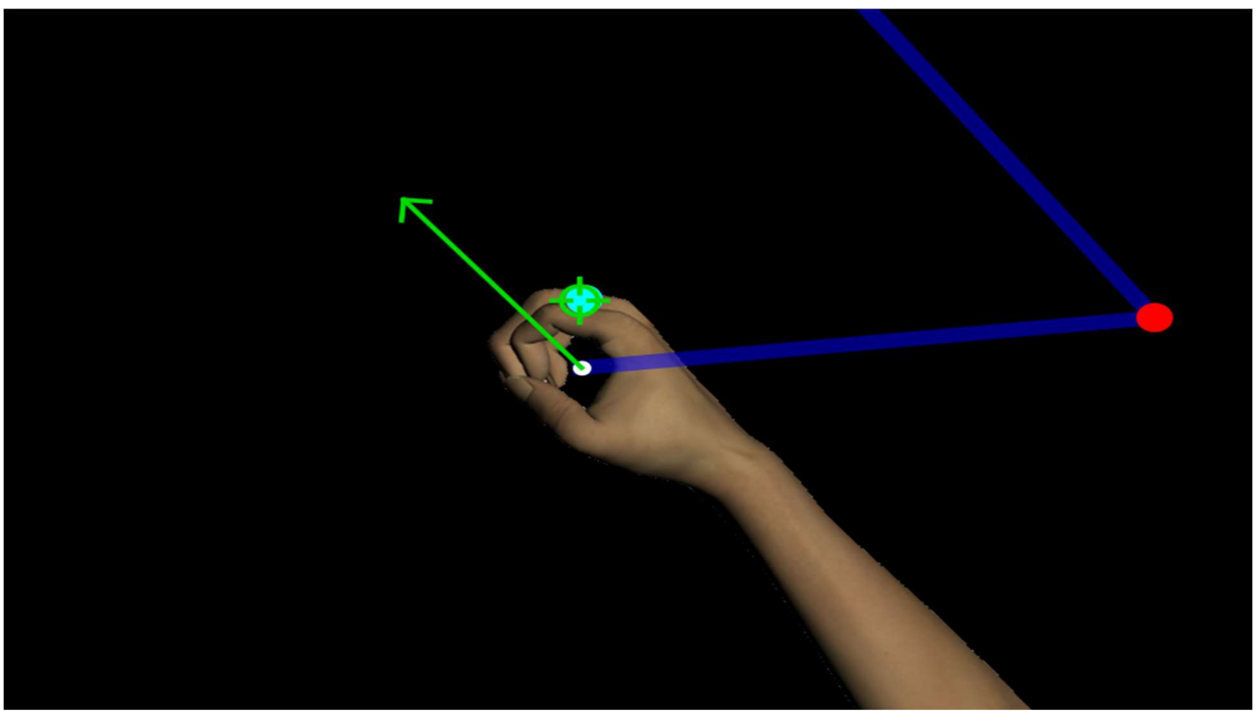


Figure 2. Dexterit-E workspace from Trial 3 of demo_1.kinarm. The participants gaze (green crosshair) is locked to the target in smooth pursuit as the right hand moves to intercept it.

respectively). The participants were instructed to reach the target in all conditions. *Figure 2* is a picture of the demo file in the Dexterit-E Explorer visualization tool.

In this demonstration, the X location of the hand, eye, and target will all be exported and used. The target's X and Y position were fed into the analog outputs of the trial and can be found in analog output channels 15 and 16. Below is a picture of those data streams within Trial 3 of [demo_1.kinarm] on Dexterit-E Explorer.

In *Figure 3*, we can see the X position of the hand (color), eye (color), and target (color) during the third trial (TP 5) aligned to an event called 'TARGET_ON' (target onset). Let's use this visualization as a bridge to connect our understanding of a KINARM experiment to the GECKO software. Continue to 'Opening the Software' to begin the demonstration.

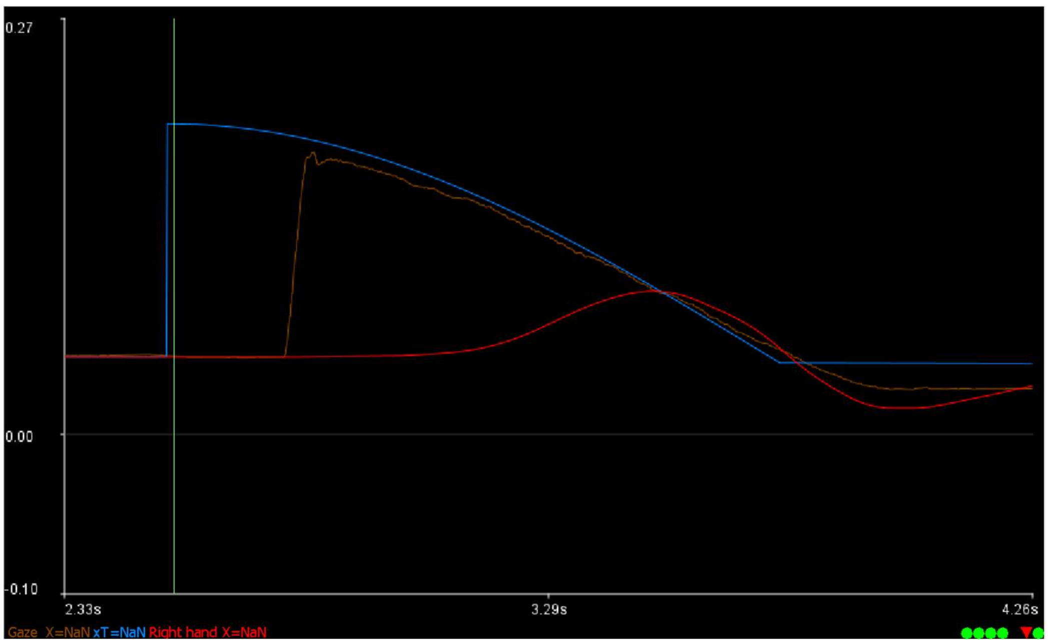


Figure 3. At the onset of target appearance (green vertical line, TARGET_ON event), the X position of the target (blue), gaze (brown), and hand (red).

Opening the Software:

To follow along with the DEMO using [demo_1.kinarm] – pay attention to these orange boxes which will narrate the process throughout the user manual…!

First, let's open GECKO, and load in our demo file [demo_1.kinarm]. We'll learn about the File Management Tab and how to save/load files.

GECKO is distributed as a standalone executable (.exe) file. To begin using the software, download the executable file and double-click it to launch the application. This user manual is based on version 1. 5. 0.

The GECKO interface is designed to work with .kinarm data files. Ensure that you have access to a directory containing the required .kinarm files generated from your experiment before starting.

1. Download the DEMO file [demo_1.kinarm].
2. Download the GECKO .exe file.
3. Double-click the executable to launch the software.

Homepage Overview:

The GECKO homepage contains several functional regions used to manage files, inspect data channels, perform computations, and configure export settings. Upon opening GECKO, the screen shown in *Figure 4* will appear.

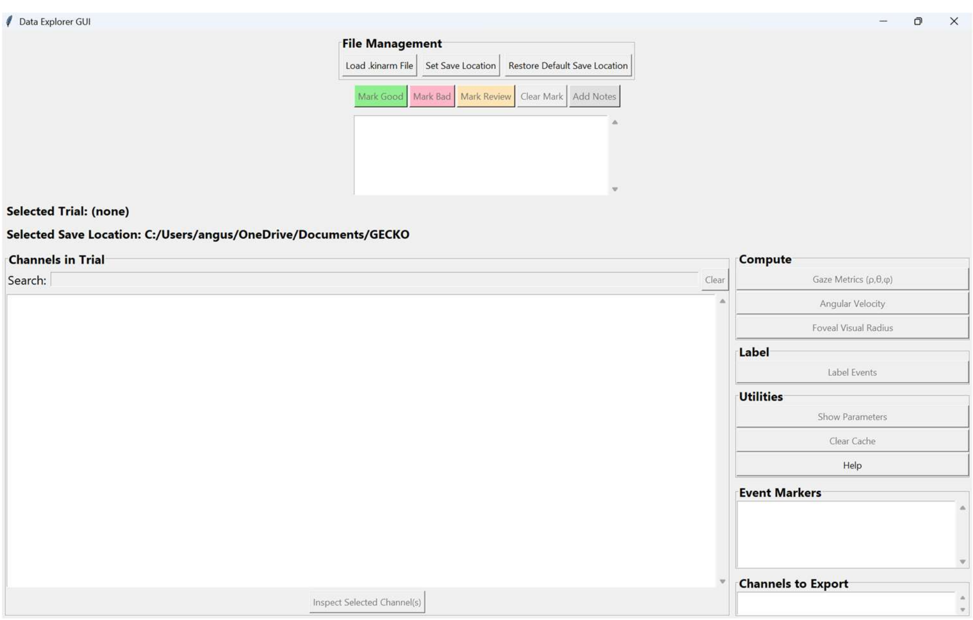


Figure 4. Homepage upon opening GECKO.exe.

File Management:

The File Management section allows users to load .kinarm files and control the working data session. Once a file is loaded, all trials contained within the file populate in the trial list window (*Figure 7*).

1. Set a save location before beginning any labeling so that exported .csv files are saved to a known directory. We will return to this location to inspect the preprocessed files later.

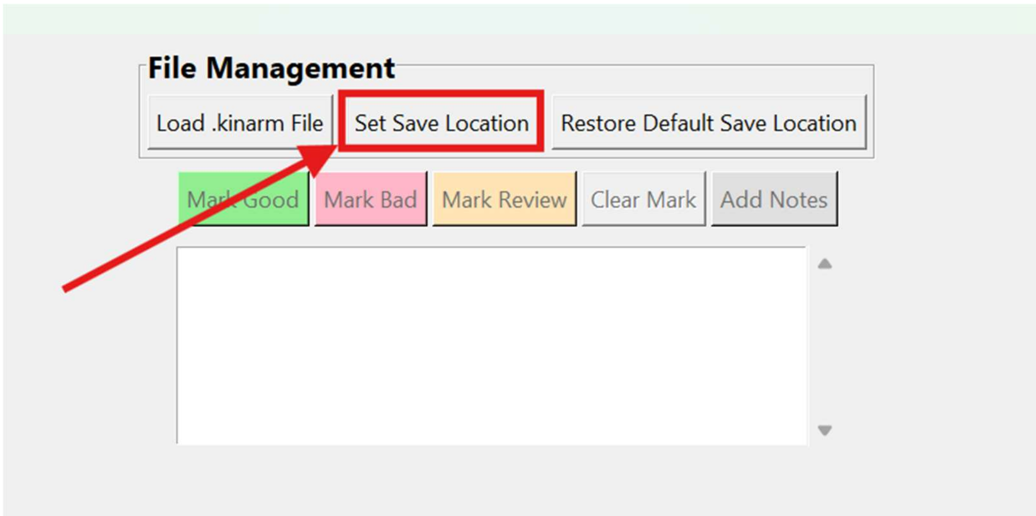


Figure 5. Set save location button on Homepage of GECKO.exe.

2. Click 'Load .kinarm File'.

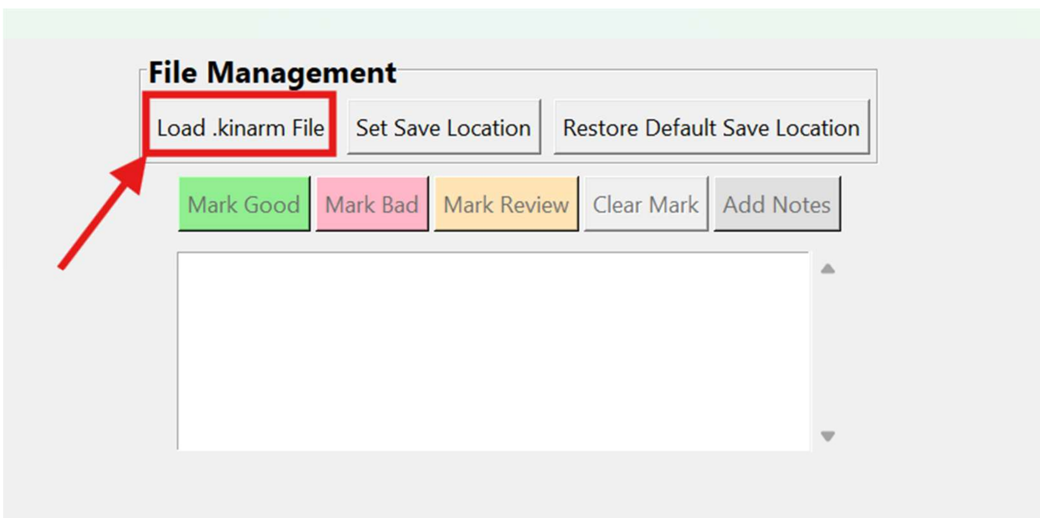


Figure 6. Button used to load kinarm files.

3. Navigate to the directory containing the desired .kinarm file and select it (find [demo_1.kinarm] in your Downloads). 'Restore Default Save Location' will change the save location back to a folder titled 'gaze_labels' on the desktop.

After loading a .kinarm file, the white window located beneath the 'File Management' section will populate with all trials contained within the file.

Figure 7. Trial window that appears after loading demo_1.kinarm. Trial 3 is selected

Each trial is named using a structured convention. The first number represents the sequential trial number within the block (displayed to the left of the period, e.g., '3.'). The second number corresponds to the Trial Protocol (TP) number. The third number indicates the count of how many times that specific trial protocol has been executed.

- A row of buttons located beneath the 'File Management' section *(Figure 8*) allows users to manage trial notes and quality flags. Users may mark trials as 'Good', 'Bad', or 'Review'. Notes can also be created for any trial and saved as .txt files to the designated save location. Existing notes/marks may also be deleted using the provided controls.

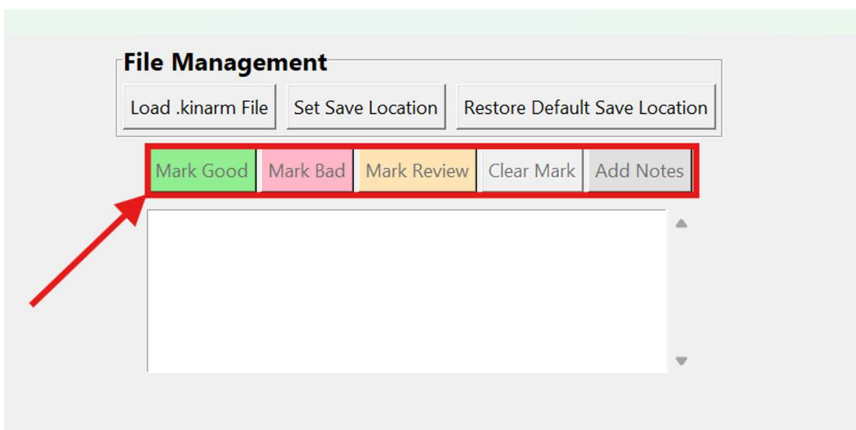


Figure 8. Marking buttons to annotate Trials within GECKO.

Selecting a trial from this list will populate all available channels within the 'Channel Inspection' and export configuration sections of the interface.

Inspect Channels / Data Visualization

In the trial window, select Trial 3 (it will become highlighted as seen in *Figure 7*). The Inspect Channels function is not a necessity to processing, but let's check it out to see how it functions in GECKO. Select the 'Gaze X' channel and click 'Inspect Selected Channel(s)'.

The channel inspection interface allows the user to visualize any data channel contained within the loaded .kinarm file. Channels can be searched using the search bar and selected for inspection. Multiple channels may be selected simultaneously.

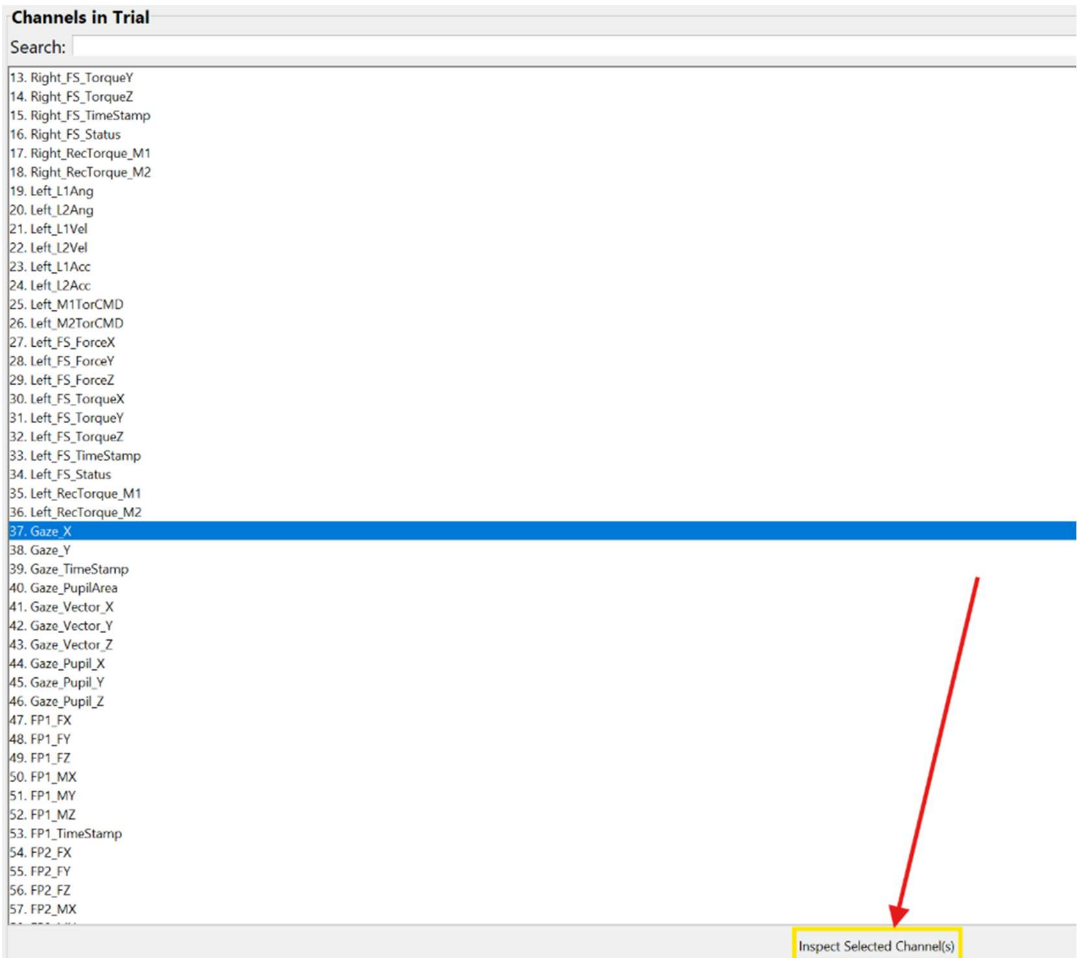


Figure 9. 'Gaze X' channel is selected from Trial 3 in the visualization tool of GECKO. Clicking the 'Inspect Selected Channel(s) button will open Figure 10 below.

Clicking 'Inspect Channel' will display the selected channels frame-by-frame within the visualization window.

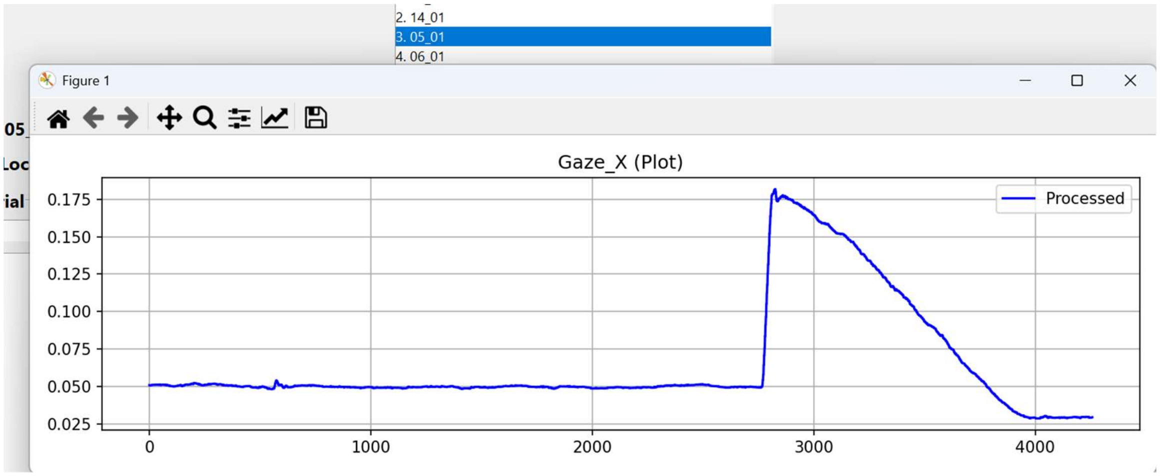


Figure 10. Gaze X channel from Trial 3 of demo_1.kinarm in GECKO appears in a separate window from the GECKO homepage after pressing 'Inspect Selected Channel(s).

As seen in *Figure 10*, the channels from the .kinarm file are saved and viewable in GECKO similar to the Dexterit-E Explorer (Figure 11).

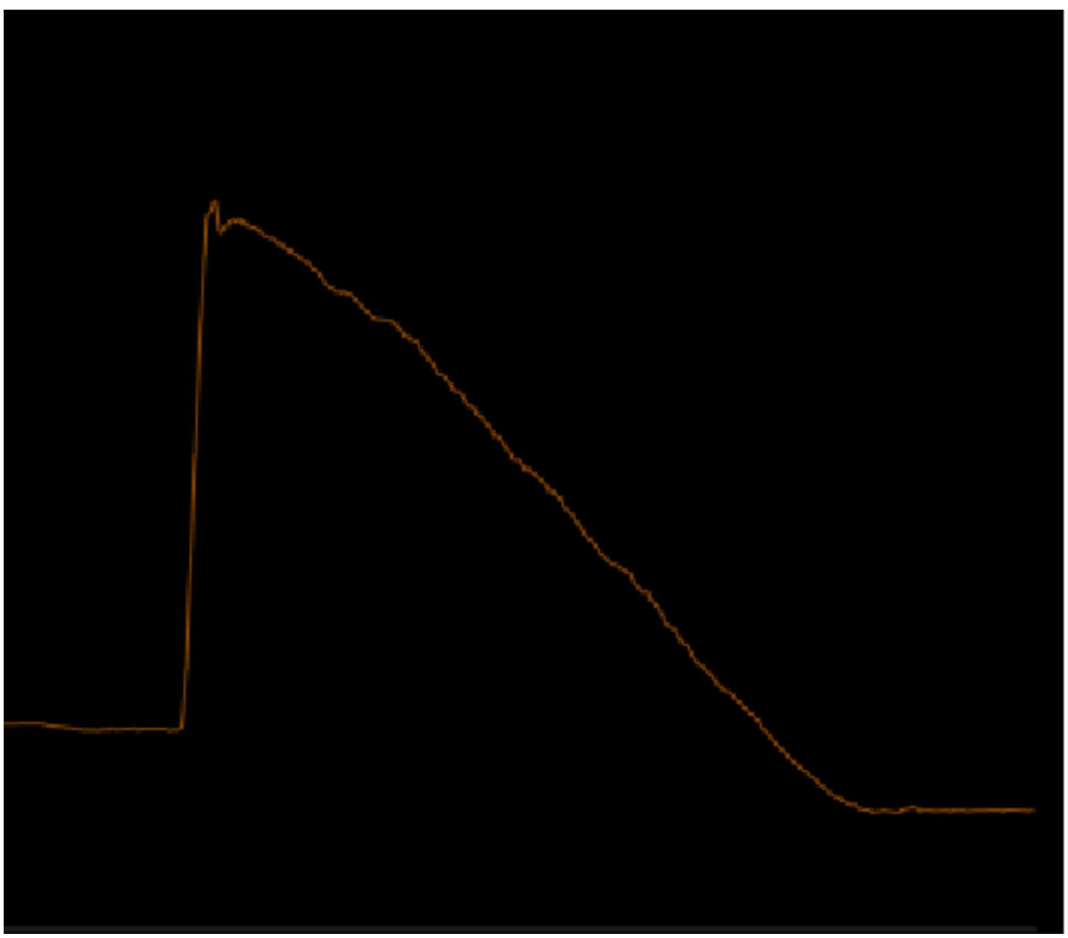


Figure 11. Gaze X channel from Trial 3 of demo_1.kinarm in the Dexterit-E Explorer.

Channels to Export After Labeling

We have seen how to load .kinarm files, set a save location for the processed trials and figured out how to visualize the data. Let's start with the processing flow by selecting which Channels to Export after labelling.

These are channels that will appear in every CSV file saved for each trial. The events in the trial are also seen in this window (*Figure 12, bottom)* and should be selected if important to paradigm (for this demo, target onset is important to calculating reaction time so TARGET_ON will be selected). See 'Saved Data' for more on this.

Below the 'Event Markers' section is the 'Channels to Export' panel. All derived and analog channels available for the selected trial populate in this section. ***These channels mirror the channels available in the inspection window; however, the channels selected in this panel will be exported to the output CSV file.***

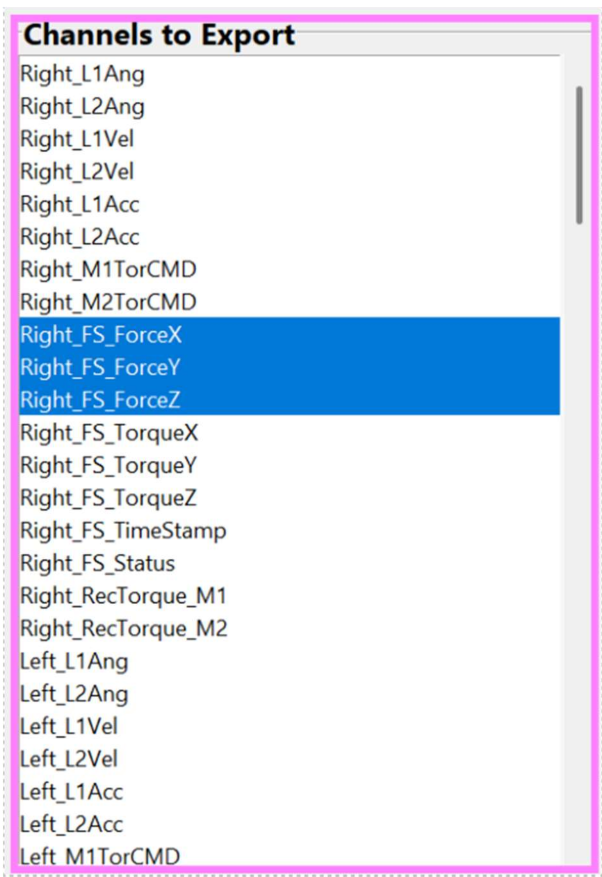


Figure 12. Some of the channels available to be selected/exported after labelling. Clicking a channel will highlight it blue and indicate it will be exported.

In the exported CSV file, Column A contains the frame number. Each selected channel will appear as a labeled column of data. Channel selections persist across trials and across newly loaded .kinarm files (see ‘Saved Data’).

<u>Event Markers for Labelling:</u>

We now know how to select channels for export in GECKO. Let’s set up the labelling part of the processing by selecting events to appear in our labelling window…! This is different than selecting them to be exported – here they are only used to help classify gaze events.

Events that dictate the target’s appearance and stoppage of its motion will be helpful when labelling the gaze. Context of target movement will be crucial to properly labelling smooth pursuit eye movements…!

The Event Markers dropdown menu displays all events associated with the currently uploaded .kinarm file. Each selected event is visualized in the gaze inspection window (see in ‘Labeling’ section, and *Figure 13* below) as a vertical line corresponding to the frame at which the event occurred.

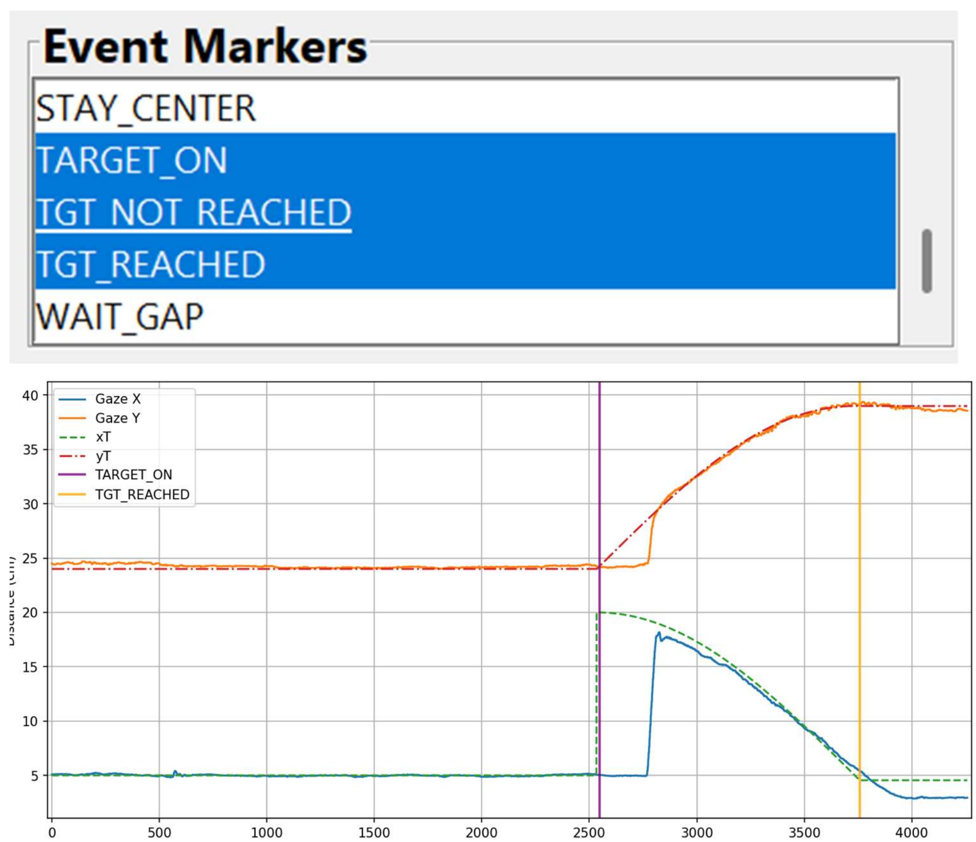


Figure 13. Event Markers selector and picture of ‘TARGET_ON’ and ‘TGT_REACHED’ events are in the Gaze Labelling window. Trial 3 of [demo_1.kinarm]

Labelling

We have selected important channels to be in our preprocessed files and chose which events should be seen while labelling gaze events.

This section goes through the labelling of gaze events and interpolation of gaze data. This is the main function of GECKO – full and temporally accurate control of gaze event classification…!

A few labeling preferences must be set before labeling can start…

On the GECKO Homepage, selecting 'Label Events' (*Figure 14*) will open the labeling window. Opening labelling for the first time will cue the user to select an order of events to label, seen in *Figure 15*. Order selections will save across .kinarm files or until the user clears the selection cache (see 'Clear Cache' section in *Other Functions*). The main labeling window is seen in *Figures 18 and 19*.

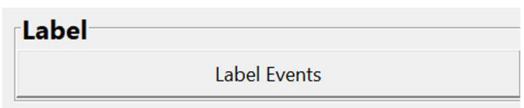


Figure 14. Label Events button. Selecting this button will begin the labeling and CSV output saving process.

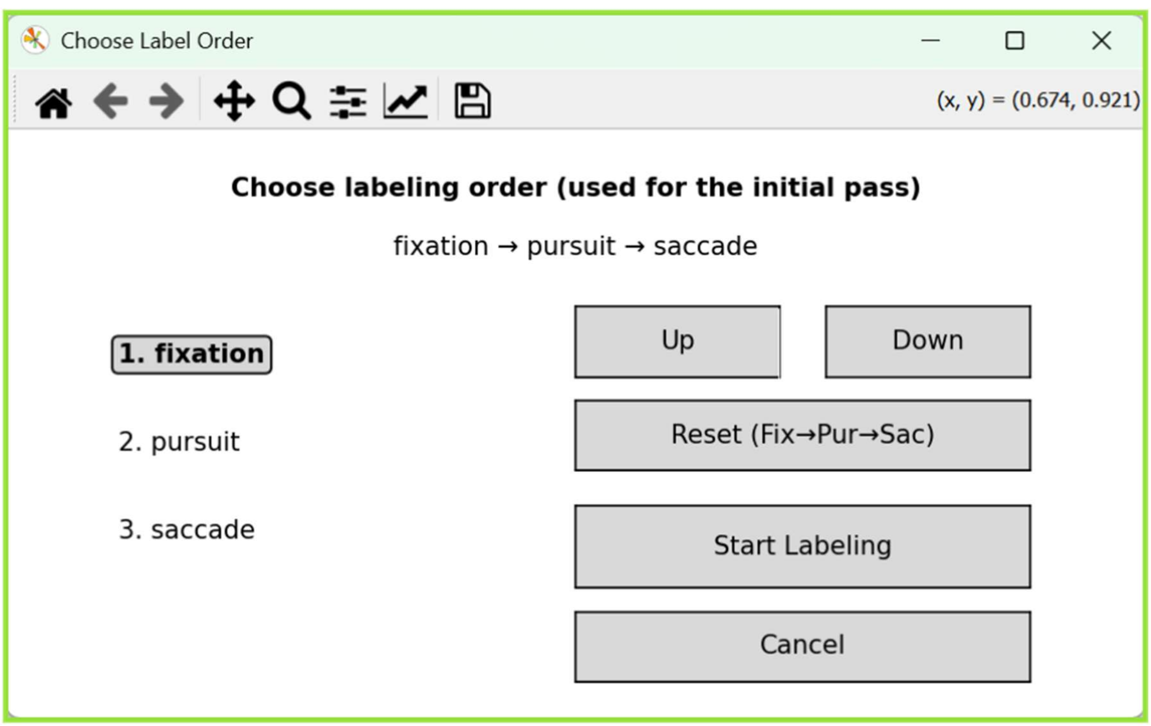


Figure 15. Order of gaze events to label. Customizable to researchers' needs

In this window, the user can select the order of gaze events to be labeled/classified within the trial. After choosing an order to label in, users will be cued to choose channels to appear in the labeling interface (*Figure 16*). Similar to the 'Event Markers' (*Figure 13*), these channels will appear in the labeling window, **<u>but selection here does not mean it will be exported to the saved csv file.</u>** See *Figure 12* and 'Select Channels to Export After Labeling' for selecting the data to be saved for each trial in the CSV file after labeling is finished. Once an order is selected (or if a previously selected order has been saved) and channels for labeling are decided, the gaze labeling interface will open (*Figures 18, 19*).

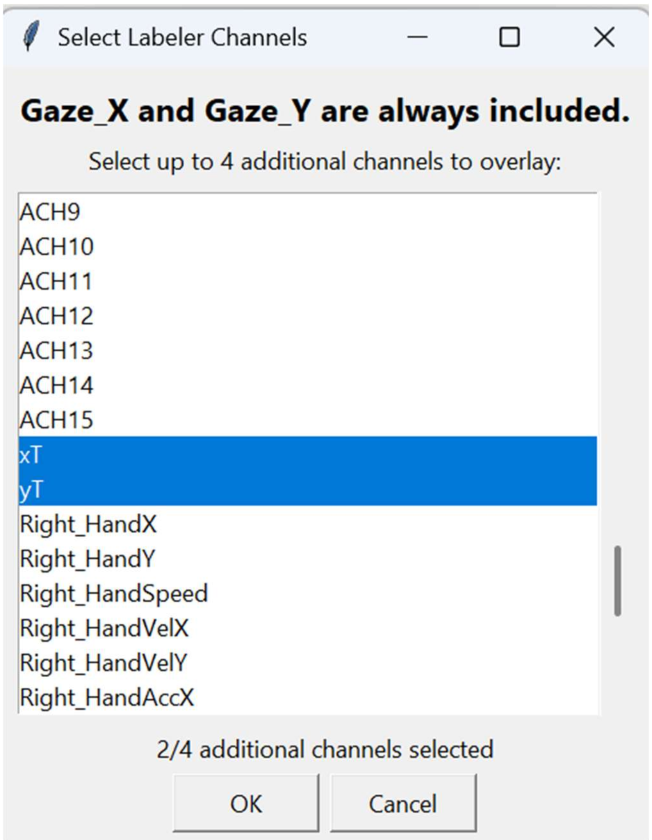


Figure 16. Channels to appear when labeling. In [demo_1.kinarm] the X and Y position of the target are analog inputs 'xT' and 'yT' respectively. These are selected above, and further images of the labeling window will follow this selection preference.

<u>Interpolation:</u>

If gaps exist in the gaze data streams (Gaze X and Gaze Y) that exceed 50 frames in duration, an interpolation window will automatically appear ***<u>before labelling can begin.</u>***

GECKO automatically interpolates gaps of 50 frames or fewer without user intervention. Larger gaps require user selection of an interpolation method, and the interpolation window shown below (*Figure 17*) will appear before labeling.

It will appear twice, once for each gaze data stream (since the gaze X or Y will not be missing without the other missing as well).

Two interpolation methods are available: Linear interpolation and Saccadic interpolation.

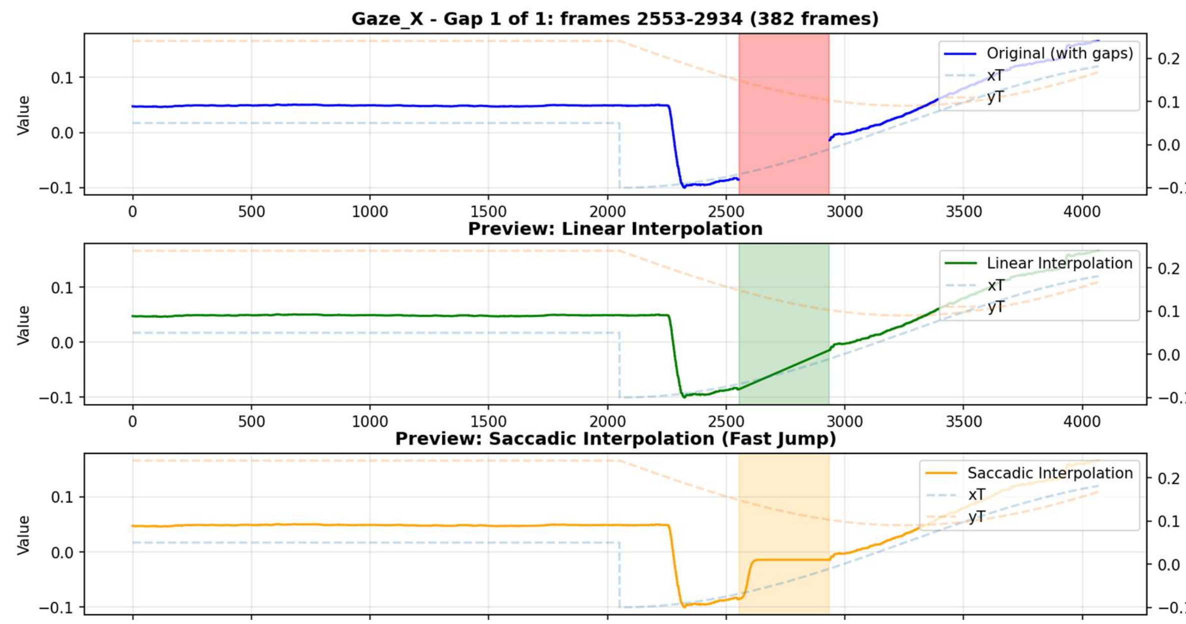


Figure 17. Interpolation window for Trial 4 of [demo_1.kinarm]. The three rows show the data without interpolation (blue, top), with linear interpolation (green, middle), or saccadic interpolation (orange, bottom).

Linear interpolation draws a straight line between the final valid sample before the gap and the first valid sample after the gap. Researchers may prefer this method when the missing data occurs during smooth pursuit, where gaze position is expected to change gradually (such is the case in *Figure 17*).

Saccadic interpolation creates a step-like transition using two straight segments: one vertical and one horizontal (in that order, respectively). This method approximates a rapid change in gaze position and may be appropriate when the gap occurs during a large translation of gaze or a potential saccadic eye movement.

You may also choose to leave the gap as 'NaN' values if you prefer to retain the missing data segment instead of interpolating it. If more than one large gap exists, the user will be cued to choose an interpolation type for each gap for each data stream.

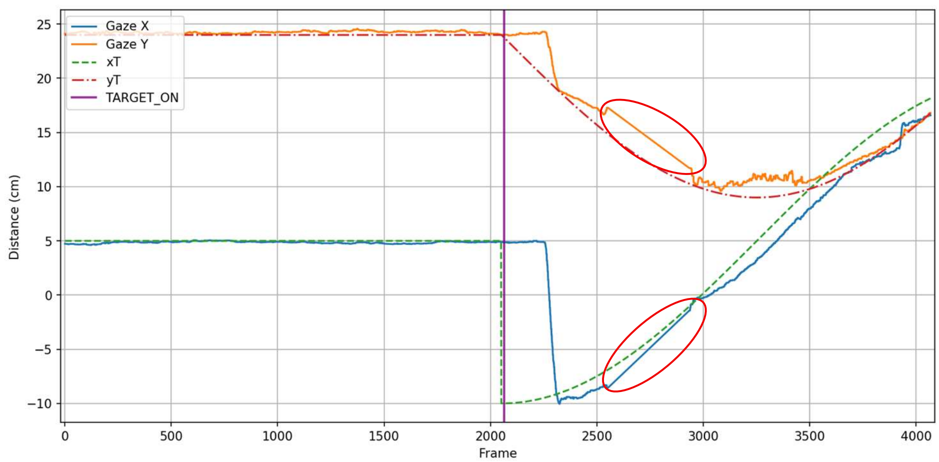


Figure 18. Gaze labelling window after interpolating for [demo_1.kinarm] Trial 4. Interpolated areas are shown in the red circles. Here, a linear interpolation was chosen to fit smooth pursuit behavior.

The Event Markers and Channels to appear during labeling are preferences that will save after being set, and can be changed by using the 'Clear Cache' button (see 'Other Functions').

Interpolation can be common depending on the quality of gaze data that was collected. After interpolating the gaze in Trial 4 of the demo data, we are ready to classify the gaze during the trial…!

Selecting the 'Label Events' button will open the gaze labelling interface right away if no gaps exist, or after the interpolation is finished, and if the event order is already dictated.

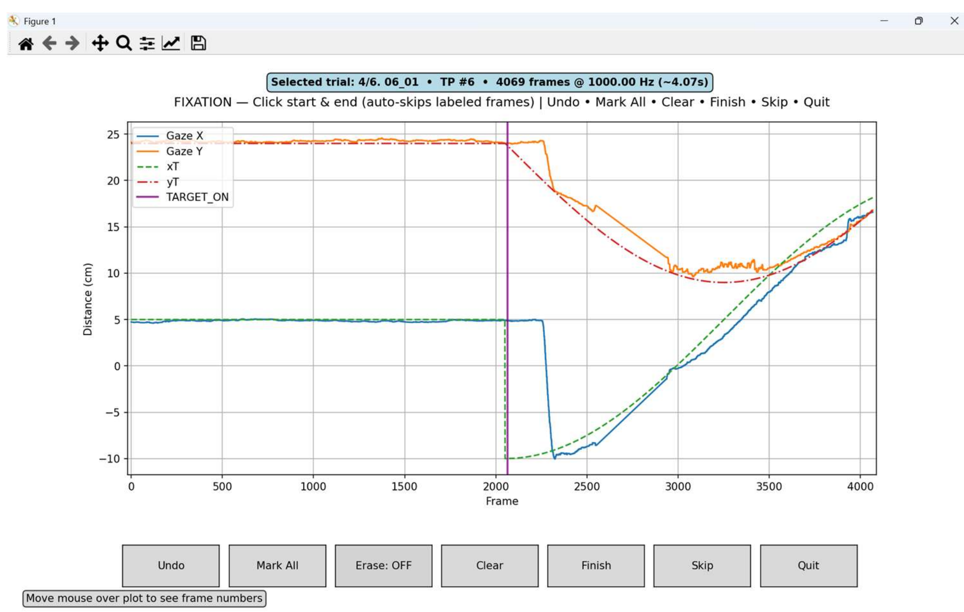
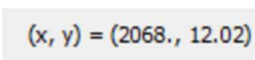


Figure 19. Full gaze labelling window of Trial 4 from [demo_1.kinarm].

The labeling interface displays gaze X and gaze Y position within the virtual display space across the duration of the trial. In addition to the gaze signals, vertical lines will appear corresponding to the events selected in the 'Event Marker' section (*Figure 13*), and other channels will appear if they were selected in the 'Channels for Labeling' section (*Figure 16*).

The X-axis represents the frames of the trial (*Figure 19*). The Y-axis represents position in centimeters within the virtual display space.

At the top of the interface, trial information is displayed along with the gaze event (Fixation, Pursuit, or Saccade) currently being labeled (Fixation, in *Figure 19*). In the gaze labeling interface, the current frame number under the mouse cursor is displayed in the bottom-left corner of the window. The top-right corner displays the frame number as well as the Y-axis value corresponding to the current cursor/mouse position (in cm).



To label/classify an event, click on the graph at the frame where the event begins. Clicking a second time will define the end of the gaze event interval. The region between the two clicks will be highlighted as the event currently being labeled. Here, a saccade was labeled (red highlighted area, *Figure 20*).

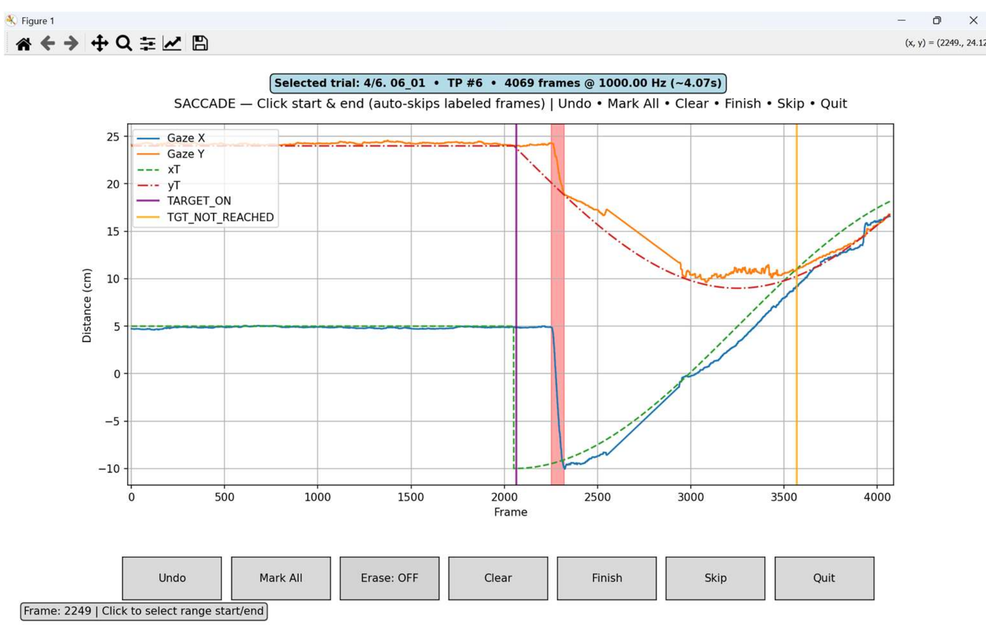


Figure 20. Gaze labeling window after a saccade has been labeled, shown as the red highlighted area.

You can label as many events of the same class as needed by continuing this process. Clicking on already classified/highlighted regions will fill the current gaze event to the border of that already highlighted section(s). A small amount of padding is displayed before frame to make it easier for the user to select events at the beginning.

The following controls are available during labeling:

Undo – removes the most recent event placement.

Mark All – labels all currently unlabeled segments of the trial as the active event type.

Erase – selecting two places on the plot by clicking will erase the labelling in the region between the 2 clicks.

Clear – removes all intervals of the current event type.

Finish – saves the labeled intervals for the current event type and advances to the next event.

Skip – advances to the next event in the labeling order without placing a label.

Quit – leaves the gaze labelling window without saving.

Label the events and use ‘Finish’ to navigate through all 3 main event classifications (Label all the saccades, then pressing ‘Finish’ will bring you to ‘Pursuit’. After labeling the smooth pursuits, pressing ‘Finish’ will bring you to ‘Fixation’). After completing labeling for each gaze event type (saccade, pursuit, and fixation), the Summary interface will appear (*Figure 21*). It shows the entire trial, all selected ‘Events Markers’ (vertical bars) and data streams, as classified gaze events.

<u>Green for Fixation, Red for Saccade, and Purple for Smooth Pursuit.</u>

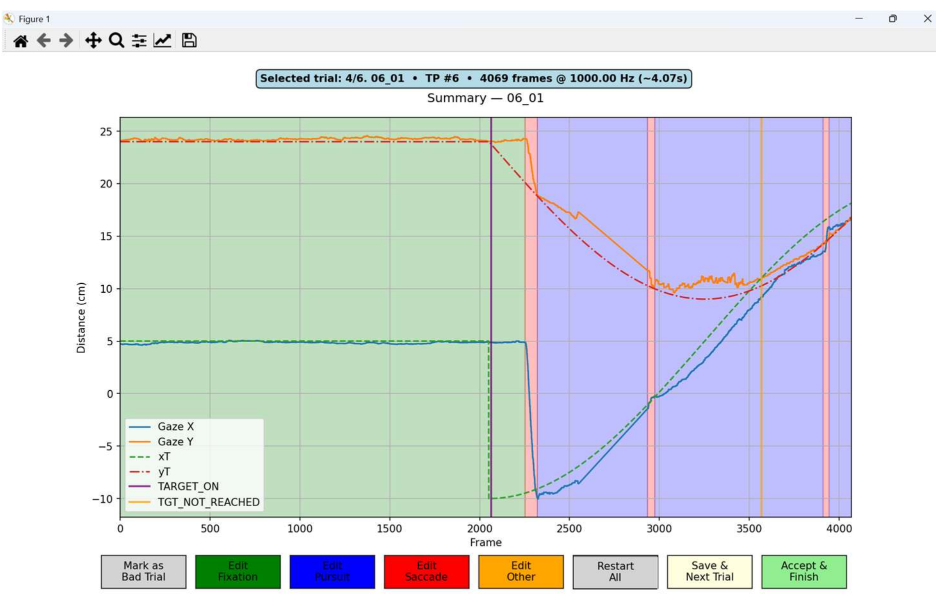


Figure 21. Summary page after labelling Trial 4 of [demo_1.kinarm].

This interface allows the user to review the labeled intervals and adjust if necessary (e.g. ‘Edit Fixation’ will allow user to remove one or all already labeled [highlighted green area, in the case of fixation], sections and replace as needed).

From this page, the user may also mark the entire trial as 'Bad'. An example of this button being pressed for a bad gaze trial is shown below:

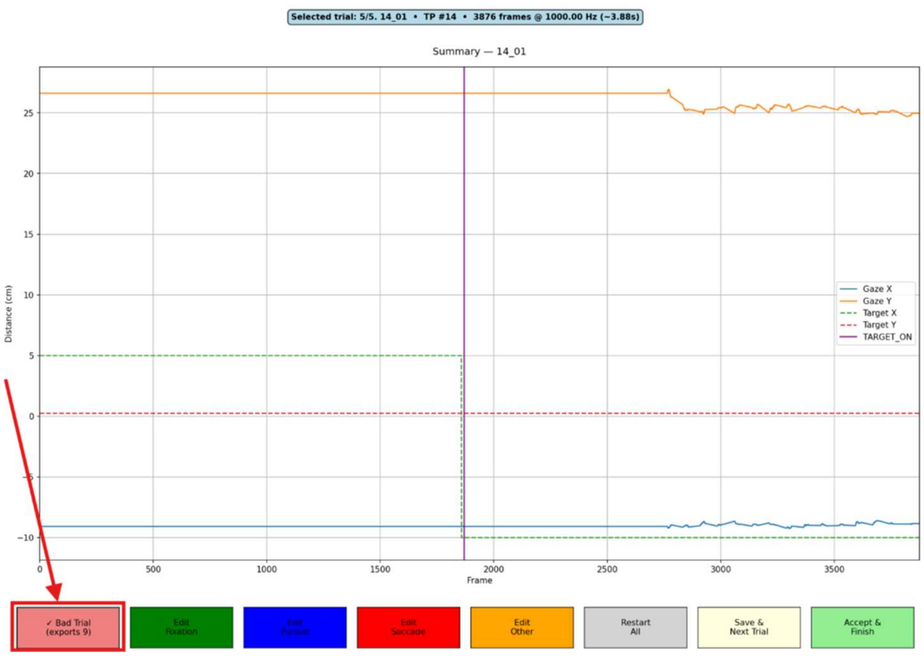


Figure 22. ‘Bad trial’ where gaze signals were not usable.

Details on how bad trials are handled during export are described in the ‘Saved Data’ section of this manual (*Figure 25*). Ambiguous gaze segments may also be labeled using the 'Other' classification option available on this screen (*Figure 22*). The user can restart all gaze classifications on the same trial by pressing the ‘Restart All’ button. Pressing 'Accept and Finish' generates the CSV output file for that trial and closes the gaze labeling interface.

Selecting 'Save and Next Trial' generates the CSV output file for the current trial and automatically loads the next trial in the block. This is the best method to processing all the trials in a block…!

All the same choices of channels to export, channels/events to show while labeling will persist across all trials unless changed.

Saved Data

For every processed trial, there will be 1 CSV file and 1 NPZ file created. They will all be stored in a folder titled with the name of the .kinarm file – and in the directory specified in the Set Save Location button.

We will now go over what's being saved in the CSV…!

After labeling is completed, GECKO saves all outputs to the user-defined save directory. The save directory will contain several types of files and folders representing the labeled data and session metadata. An example save location is shown below.

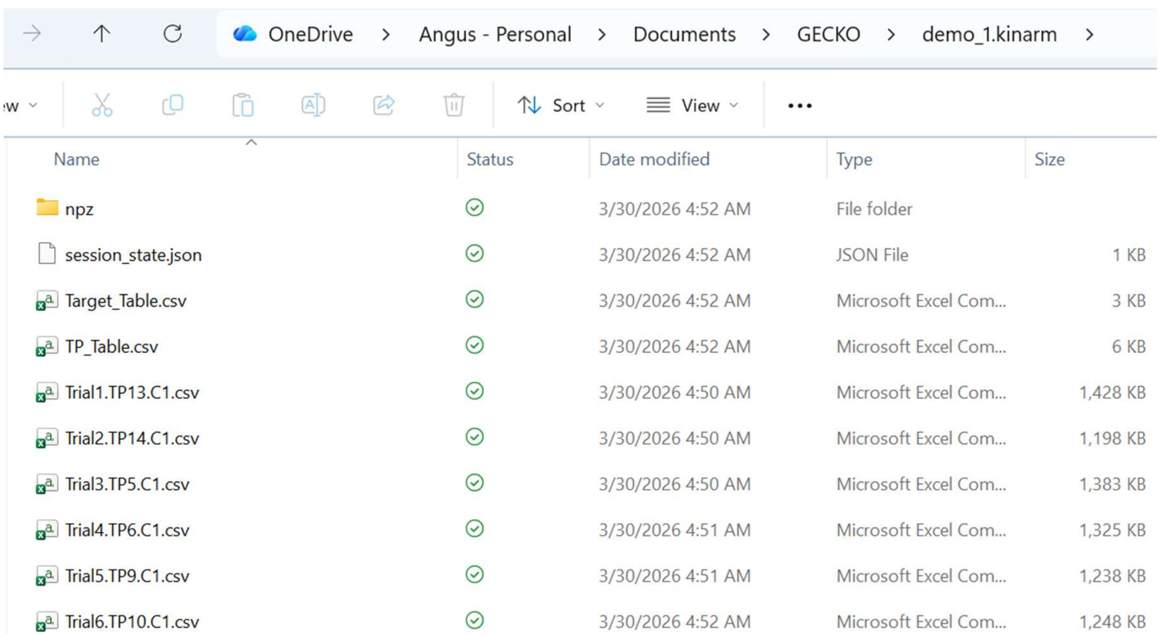


Figure 23. Example directory in the set save location. A folder containing npz files for all 6 trials, a session save file, Target and TP table from the .dtp file of the task, and CSV files for each processed trial.

NPZ Folder

A folder named 'npz' contains compressed .npz files storing the same information contained in the exported CSV files. These files are included for storage efficiency because they occupy less disk space.

Session State File

If the gaze labeller was closed before completing all trials, a session state file will appear in the save directory. This file allows the software to restore the previous session and continue labeling from where the user left off.

Target and TP Tables

CSV files containing the Target table and TP table will also be exported. These tables correspond to the same information contained in the .dtp file used during data collection.

Trial CSV Files

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Frame	Gaze_Events	Right_FS_ForceX	Right_FS_ForceY	Right_FS_ForceZ	Gaze_X	Gaze_Y	Gaze_TimeStamp	TRICEPS E	DELTOID E	PECTORAL	xT	yT	Right_Han	Right_Han	Right_Han	Right_Han	Rho (Dista	Theta (Azir	Phi (Elevat	Angular_V	TGT_REACHED
2	0	3	-0.111450411	0.41935423	-17.78561211	0.050975	0.243957	25.00600052	-0.00014	0.00796	-0.00467	0.05	0.24	0.04683	0.236158	0.024021	0.024384	0.319552	1.364809	0.894543	27.01661	3402
3	1	3	-0.113312982	0.419474959	-17.8381176	0.050999	0.244008	25.00799942	-0.00078	0.008445	-0.00483	0.05	0.24	0.046854	0.236179	0.024537	0.022902	0.319595	1.364757	0.894649	9.612806	
4	2	3	-0.109624915	0.420515567	-17.86156464	0.050999	0.244008	25.00899982	-0.00062	0.008445	-0.00483	0.05	0.24	0.04688	0.236209	0.025056	0.02142	0.319595	1.364757	0.894649	4.333672	
5	3	3	-0.112405963	0.426944166	-17.88582039	0.050999	0.244008	25.01000023	-0.00046	0.008931	-0.00418	0.05	0.24	0.046908	0.236223	0.025575	0.019937	0.319595	1.364757	0.894649	14.05774	
6	4	3	-0.131981879	0.480161756	-17.88536453	0.050999	0.244008	25.01099968	-0.00062	0.008607	-0.00499	0.05	0.24	0.04694	0.236229	0.026096	0.018458	0.319595	1.364757	0.894649	20.10899	
7	5	3	-0.482861668	1.411822915	-17.98911095	0.051004	0.243594	25.01199913	-0.00078	0.008284	-0.00499	0.05	0.24	0.046974	0.236232	0.026615	0.016983	0.319279	1.364395	0.893856	22.42698	
8	6	3	-0.955002427	2.384394169	-17.85333443	0.051004	0.243594	25.01299953	-0.0003	0.008607	-0.00418	0.05	0.24	0.047009	0.23624	0.027135	0.015514	0.319279	1.364395	0.893856	20.58044	
9	7	3	-1.185594559	3.037736654	-17.38814735	0.050988	0.243172	25.01399994	-0.0003	0.008607	-0.00435	0.05	0.24	0.047047	0.236228	0.027652	0.014053	0.318955	1.364112	0.893038	14.19356	
10	8	3	-1.309082747	3.402053356	-17.25736618	0.050988	0.243172	25.01500034	-0.00095	0.008445	-0.00451	0.05	0.24	0.047089	0.236214	0.028168	0.012602	0.318955	1.364112	0.893038	7.143409	
11	9	3	-1.726661801	4.026605606	-17.28411484	0.050949	0.243305	25.01600075	-0.00062	0.008607	-0.00451	0.05	0.24	0.047127	0.236202	0.028681	0.011163	0.319051	1.364377	0.89328	6.219421	
12	10	3	-2.056256294	4.265402794	-17.62298965	0.050949	0.243305	25.0170002	-0.00062	0.008769	-0.00435	0.05	0.24	0.047158	0.236197	0.029191	0.009736	0.319051	1.364377	0.89328	6.48112	
13	11	3	-2.278210878	4.592308521	-17.9325428	0.050851	0.243305	25.01799965	-0.0003	0.008284	-0.00451	0.05	0.24	0.047202	0.236184	0.029696	0.008323	0.319035	1.36476	0.893241	5.814243	
14	12	3	-2.327844143	4.70676136	-17.77540779	0.050851	0.243305	25.01900005	-0.0003	0.008284	-0.00451	0.05	0.24	0.047238	0.236171	0.030197	0.006926	0.319035	1.36476	0.893241	4.438817	
15	13	3	-2.249094963	4.618216515	-17.35075188	0.050951	0.243345	25.02000046	-0.00046	0.008284	-0.00483	0.05	0.24	0.047281	0.236154	0.030694	0.005547	0.319081	1.3644	0.893356	5.517587	

Figure 24. Screenshot of a CSV file saved from GECKO. The 'Gaze_Events' column contains the gaze event classification at every frame that was completed in GECKO. Any channel/event selected in 'Channels to Export' will populate as columns 'C' and on.

Above is an example of one of the trials CSV files being opened. Each processed trial will produce a CSV file. The file naming convention follows the same structure as the trial list displayed in the GECKO Homepage: trial number, Trial Protocol (TP) number, and the count of that TP execution.

The gaze labeller exports a column (Column B) aligned to the frames (Column A) of the trial representing gaze event classification.

The classification values are encoded as follows:

1 – Saccade

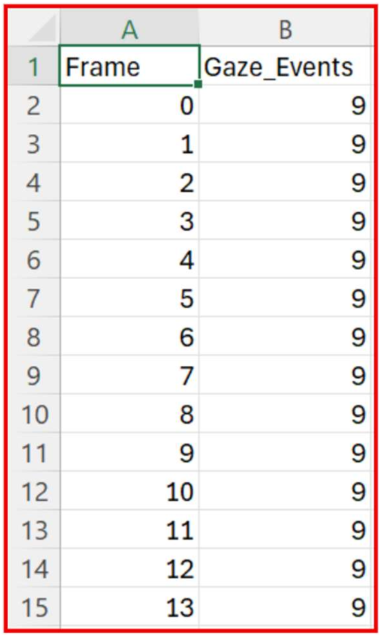
2 – Pursuit

3 – Fixation

4 – Other / ambiguous gaze segments

9 – Trial marked as bad

When a trial is marked as bad from the summary screen, the gaze classification column is filled with the value 9 across all frames.



	A	B
1	Frame	Gaze_Events
2	0	9
3	1	9
4	2	9
5	3	9
6	4	9
7	5	9
8	6	9
9	7	9
10	8	9
11	9	9
12	10	9
13	11	9
14	12	9
15	13	9

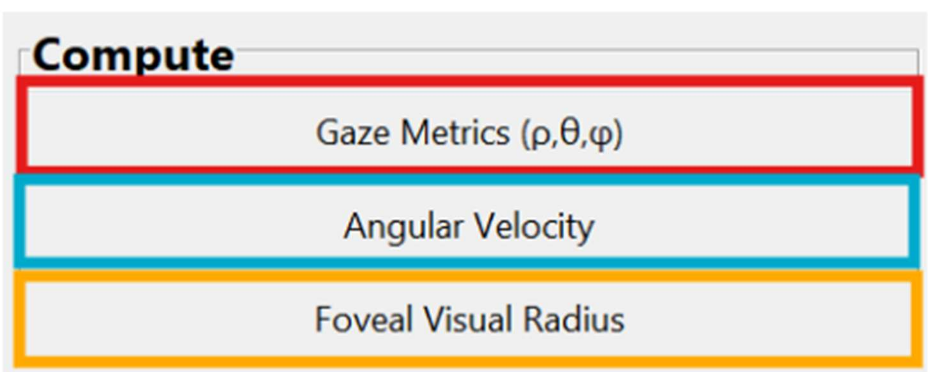
Figure 25. Screenshot of the gaze classification column of a processed trial after 'Bad Trial' was selected in the Summary page of that trial.

If additional channels were selected for export, those channels will appear as additional columns in the CSV file. When exporting gaze channels, it is recommended to also export the gaze timestamp column. The gaze timestamp sampling rate may differ from the frame rate of the KINARM robot, so including this column preserves the correct temporal alignment of gaze measurements.

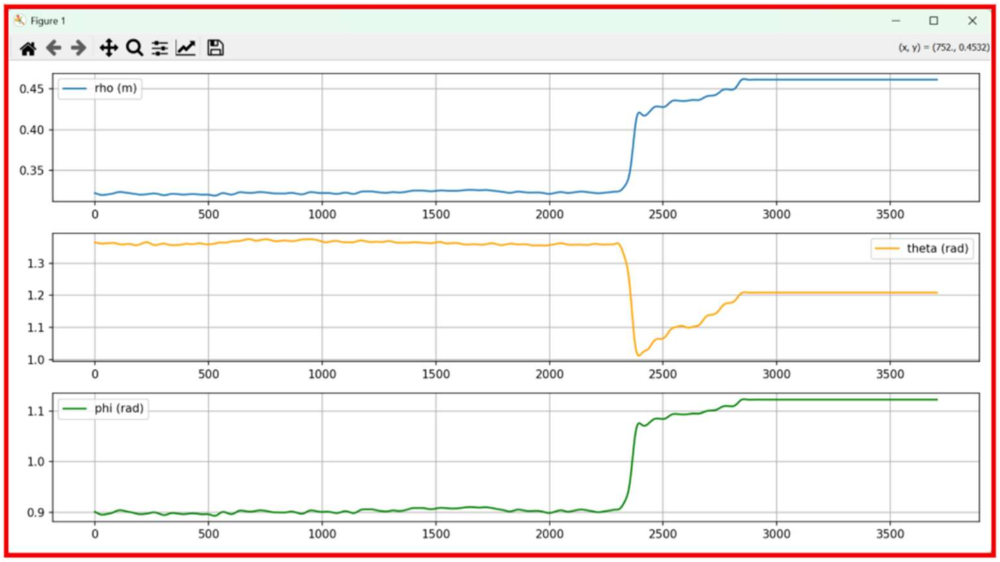
Other Functions

Computations

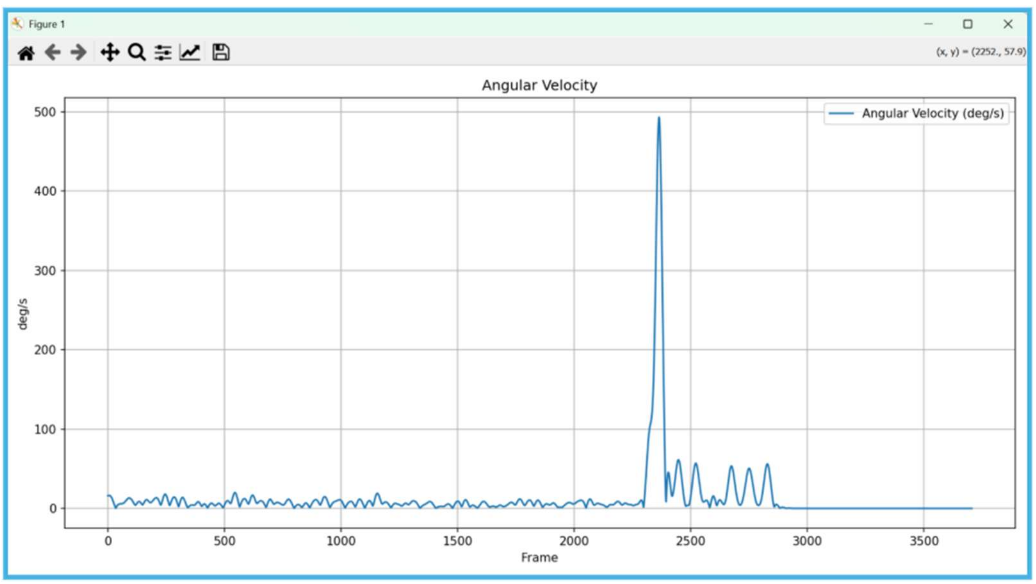
You can view computations for the currently selected trial within the Compute section.



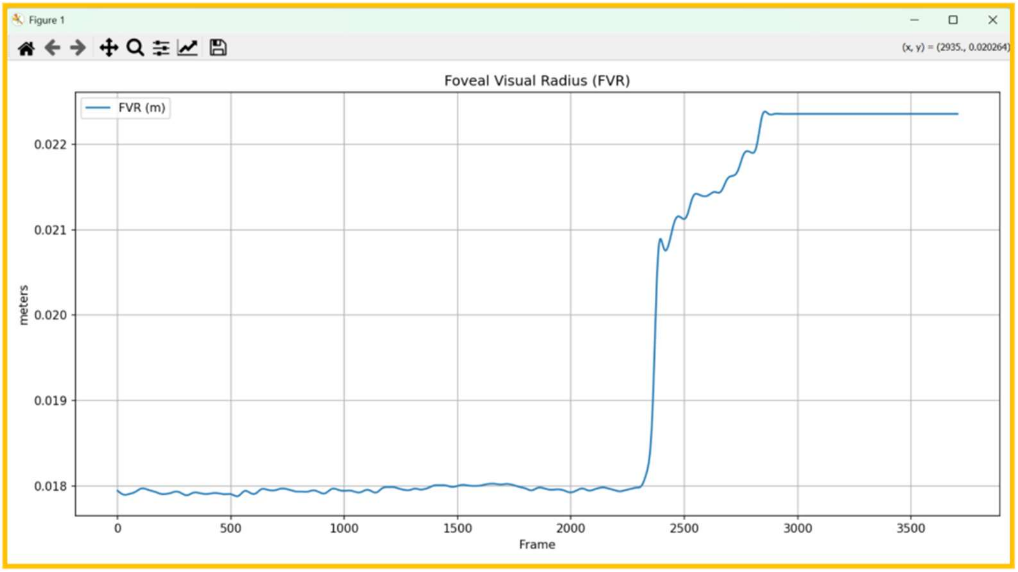
The gaze metrics displayed here are the variables used to calculate angular velocity.



[Insert citation to relevant publication. Include equations from the paper that describe how gaze metrics and angular velocity are calculated. Explain each term of the equations and how they relate to the recorded gaze data. Insert angular velocity equation from cited paper and provide explanation.]

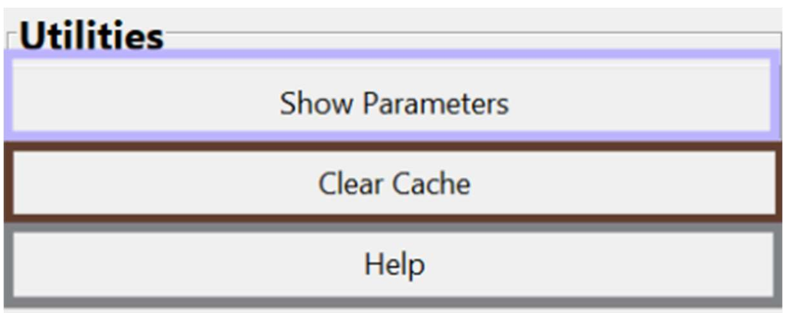


The foveal visual radius is a value computed by the Dexterit-E software and is extractable from each trial containing eye-tracking data. This value is displayed in the Compute section when available.

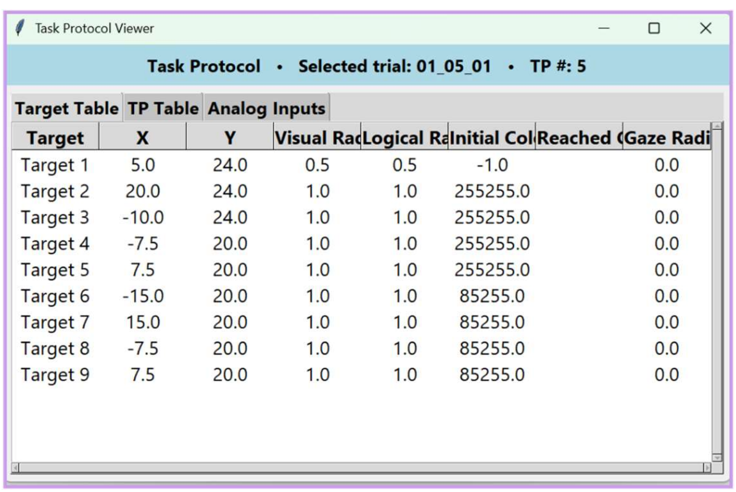


<u>Utilities</u>

The Utilities section provides several tools for session management and workflow control.

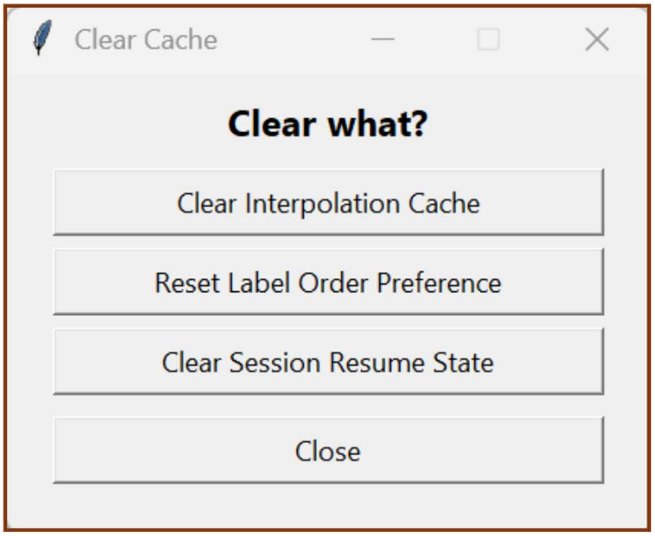


'Show Parameters' displays the Trial Protocol (TP) table associated with the loaded .kinarm file. This table is identical to the parameter table available in Dexterit-E.



Target	X	Y	Visual Rad	Logical Ra	Initial Col	Reached (	Gaze Radi
Target 1	5.0	24.0	0.5	0.5	-1.0		0.0
Target 2	20.0	24.0	1.0	1.0	255255.0		0.0
Target 3	-10.0	24.0	1.0	1.0	255255.0		0.0
Target 4	-7.5	20.0	1.0	1.0	255255.0		0.0
Target 5	7.5	20.0	1.0	1.0	255255.0		0.0
Target 6	-15.0	20.0	1.0	1.0	85255.0		0.0
Target 7	15.0	20.0	1.0	1.0	85255.0		0.0
Target 8	-7.5	20.0	1.0	1.0	85255.0		0.0
Target 9	7.5	20.0	1.0	1.0	85255.0		0.0

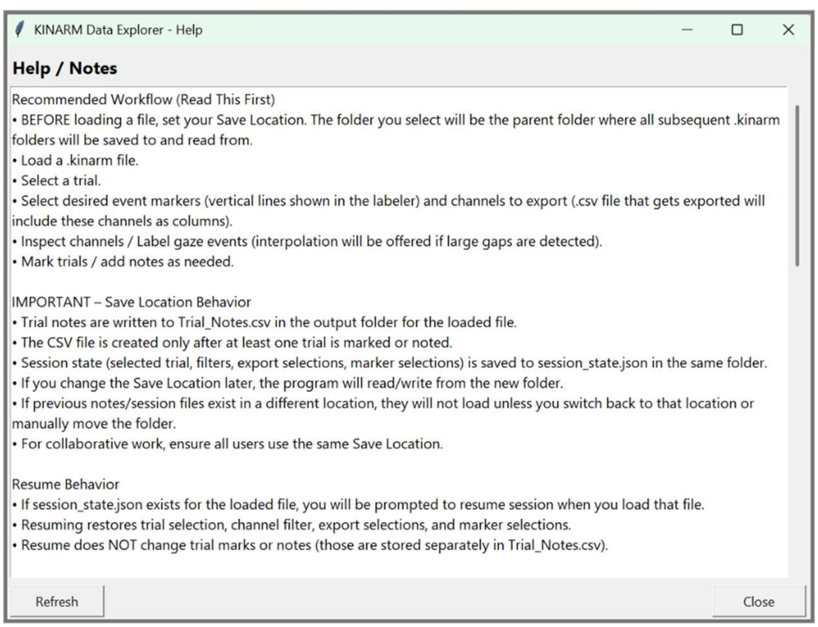
The 'Clear Cache' button allows the user to remove stored session data. Items that can be cleared include interpolation methods, labeling order preferences, and the session resume data.



Session resume allows the user to stop processing a .kinarm file midway through a session and continue later. If the software is closed and reopened, loading the same .kinarm file will restore the previous session state. This includes the current trial position, selected channels, labeling order preferences, saving location, and interpolation methods.

<u>Help / Notes</u>

The 'Help / Notes' button displays documentation from the software developer containing important considerations and additional information about GECKO.



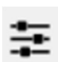
Plots:

The plotting toolbar appears on all figures displayed within the gaze labeling interface. These controls allow users to interact with and export the visualization.

This button allows the user to save the current figure.

This button enables zooming into the figure.

 This button opens tools for editing the displayed axes.

This button allows modification of plot appearance and details.

This button enables panning and scrolling across the figure or subplot.

The arrow buttons toggle between interaction modes.

The home button restores the original default view of the plot.